

EDUCATION

<i>Bachelor of Science in Information Science (Data Specialization)</i> , University of Maryland , College Park, MD	Expected December 2026
<i>Minor in Data Science</i> , University of Maryland , College Park, MD	
GPA: 3.673 Major GPA: 3.862	

SKILLS

- **Languages/Tools:** Python, R, SQL, Java, JavaScript, HTML/CSS, AWS (CDK, SDK, CLI), VSCode, Git, ArcGIS, Power BI, Tableau, Node.js, npm, Shiny, Excel, Supabase, Vercel, Docker, MySQL/PostgreSQL, OSRM/OSM, Valhalla
- **Libraries/Frameworks:** NumPy, Pandas, boto3, Pipenv, FastAPI, pscyopg, DuckDB, Geopandas, Scikit-learn, SciPy, SymPy, Matplotlib, Seaborn, Tkinter, Tidyverse, Sf, ggplot2, ggspatial, dplyr, Tidycensus, Tigris, React, Express
- **Techniques:** Agile, CI/CD, Data Processing, Statistical Modeling, Significance Testing, Machine Learning, Linear/Logistic Regression, PCA, Database Design, APIs
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Probability & Statistics, Cloud Computing

WORK EXPERIENCE

<i>Math Tutor, Calvert Hall College High School</i>	November 2019–November 2021
Community Service	Towson, MD

- Tutored immigrant students in math subjects ranging from Algebra to Calculus II; supported academic mentorship

<i>Consultant I iConsultancy Capstone</i> — GitHub	January 2026–May 2026
Southern Poverty Law Center (SPLC)	College Park, MD

- Designed and built a multi-method spatial data analysis pipeline in Python, R, and SQL that calculated polling place accessibility/distance across 203 population centers, then deployed findings into three interactive ArcGIS applications that employed directional routing, isochrone maps, and live indicators
- Presented findings to the SPLC client team in a structured final presentation alongside 10 additional peer consulting teams
- Co-presented final research findings to 240+ SPLC staff and stakeholders during an organization-wide meeting; work recognized by senior leadership and referred to SPLC HR for recruitment consideration

<i>Data Engineer Intern, CapTech</i>	June 2026–August 2026
Airline Data & Analytics	Richmond, VA

- Architected and deployed a serverless, event-driven ETL pipeline on AWS (S3 → SQS → Lambda → RDS) using Python and the CDK/CLI
- Developed EventBridge cron rules to help design and build a normalized PostgreSQL database for real-time analytics layers across 3 schemas, 20+ tables, and 30+ SQL views
- Ingested approximately 60 million rows of data while simultaneously monitoring processed outputs using AWS CloudWatch and a Power BI dashboard
- Optimized query performance speed by 70% on multi-million-row tables using targeted indexing, precomputed lookup tables, and connection-pooled database writes
- Designed data delivery layers using backend APIs to fetch and stream optimized RDS analytics outputs, ensuring low-latency rendering of necessary operational data on the frontend UI of a deployed web application.

PROJECT EXPERIENCE

<i>FIRE (First-Year Innovation and Research Experience) Project</i> — GitHub	September 2022–December 2023
Climate Computing	College Park, MD

- Conducted geospatial climate analysis on the West Antarctic Ice Sheet (WAIS) to assess its susceptibility to rising global temperatures and its implications for global sea-level rise
- Leveraged high-performance computing (Cheyenne supercomputer) to execute complex simulations using the Community Earth System Model (CESM2.1) and the Weather Research & Forecasting (WRF) Model
- Engineered data pipelines to process and visualize multi-variable climate data for presentation using Python libraries (Matplotlib, Cartopy)
- Synthesized findings into a co-authored research paper before building a poster board to present them at an academic summit; identified a correlation between reduced CO2 emissions during COVID-19 lockdowns and a short-term stabilization trend in Antarctic ice mass

<i>Database Design/Modeling Project</i> — GitHub	August 2024–December 2024
Documenting Hate	College Park, MD

- Designed a 7-table relational MySQL database from a raw ProPublica dataset to analyze hate crime trends, reducing data redundancy by normalizing the original dataset to the Third Normal Form (3NF)
- Implemented a complex Entity-Relationship Diagram (ERD) featuring one-to-many and many-to-many relationships to optimize data integrity and query efficiency
- Developed SQL views and complex queries (utilizing CTEs and subqueries) to analyze key patterns, including high-frequency keywords and geographic hotspots, enabling users to quickly identify trends across 45+ articles from 35 locations
- Cleaned and imported source data, performed forward and reverse engineering to maintain consistency between the ERD model and physical database (ETL)

<i>Exploratory Data Analysis Project</i> — Article/GitHub	September 2025–December 2025
Music Dataset	College Park, MD

- Implemented a supervised machine learning pipeline to classify songs into seven genres using combined audio signal features (e.g., danceability, energy, acousticness) and lyrical sentiment/context scores (NLP), demonstrating real-world application of feature engineering, model training, and evaluation in music data analysis
- Evaluated multiple classification models (Random Forest and Gradient Boosting) with comprehensive performance metrics (accuracy, precision, recall, F1-score, confusion matrices) and visualizations, uncovering inherent genre overlaps and model limitations due to feature ambiguity and data imbalance
- Explored unsupervised techniques (K-Means clustering and PCA) alongside supervised methods to interpret natural groupings in the dataset and reveal structural patterns in music features, offering insights into the complexity of genre boundaries and challenges for automated genre prediction

<i>Web Application Project</i> — GitHub	November 2025–December 2025
Music Explorer Program	College Park, MD

- Built a full-stack music discovery platform highlighting Black artists, genres, and cultural movements, enabling contextual exploration beyond traditional popularity-based recommendations for stakeholders
- Developed a React frontend and Node/Express backend integrating Spotify and MusicBrainz APIs, with Supabase for authentication and data persistence to support artist search, favorites, and listening trend visualizations
- Deployed a production-ready application on Vercel with clear client-server separation, demonstrating real-world architecture, API integration, and responsive UI design to support A/B testing

HONORS/ACCOMPLISHMENTS

- Dean's List, 2022–Present
- *Princeton Book Award*, 2021